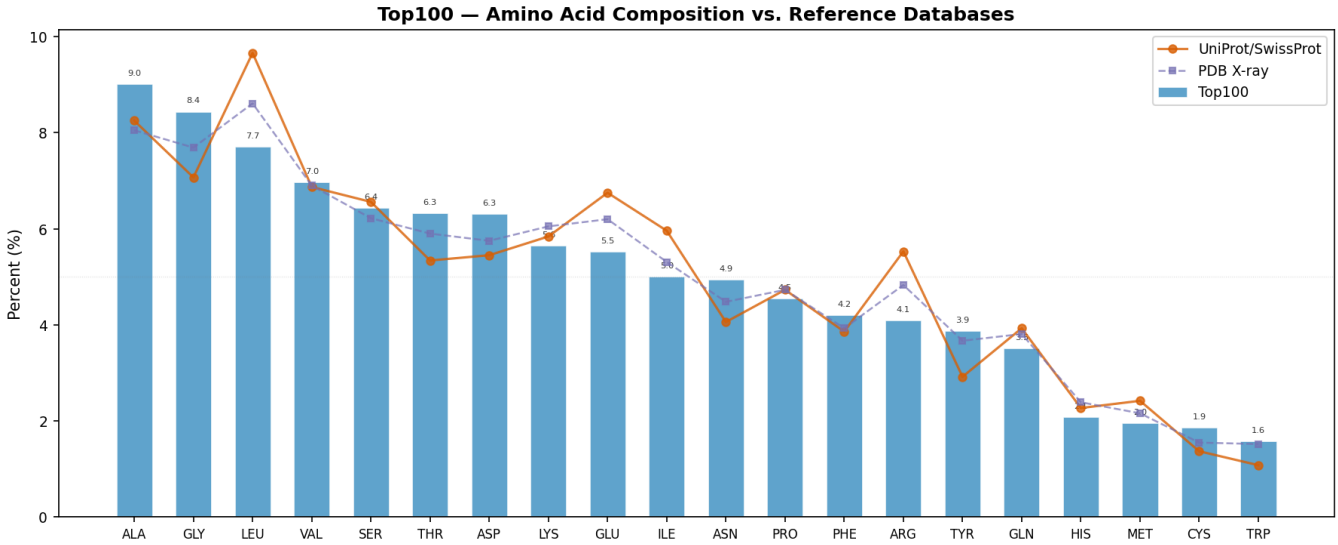
Top100 Dataset — Geometric Analysis

Residues: 17,434 | **Source:** top100_measures.jsonl | **Generated by:** pydangle-biopython v0.5.1

```
phi: 17,218  psi: 17,218  omega: 17,218  tau: 17,434
chi1: 14,380  chi2: 10,616  chi3: 6,441  chi4: 3,099
```



Residue	Count	Dataset%	UniProt%	PDB%	Δ UniProt
ALA	1,570	9.01%	8.25%	8.05%	+0.76%
GLY	1,471	8.44%	7.07%	7.69%	+1.37%
LEU	1,343	7.70%	9.66%	8.61%	-1.96%
VAL	1,215	6.97%	6.87%	6.90%	+0.10%
SER	1,122	6.44%	6.56%	6.22%	-0.12%
THR	1,102	6.32%	5.34%	5.90%	+0.98%
ASP	1,099	6.30%	5.45%	5.75%	+0.85%
LYS	985	5.65%	5.84%	6.05%	-0.19%
GLU	963	5.52%	6.75%	6.20%	-1.23%
ILE	872	5.00%	5.96%	5.31%	-0.96%
ASN	862	4.94%	4.06%	4.48%	+0.88%
PRO	792	4.54%	4.73%	4.73%	-0.19%
PHE	732	4.20%	3.86%	3.93%	+0.34%
ARG	714	4.10%	5.53%	4.83%	-1.43%
TYR	676	3.88%	2.92%	3.67%	+0.96%
GLN	613	3.52%	3.93%	3.81%	-0.41%
HIS	364	2.09%	2.27%	2.39%	-0.18%
MET	340	1.95%	2.42%	2.16%	-0.47%
CYS	325	1.86%	1.37%	1.55%	+0.49%
TRP	274	1.57%	1.08%	1.52%	+0.49%

Representativeness of Amino Acid Composition

The table above compares the Top100 amino acid frequencies against two reference databases:

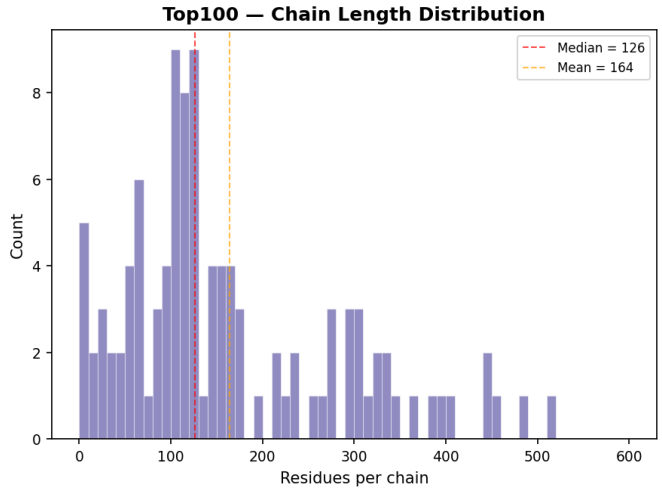
- **UniProt/SwissProt** (~2024): amino acid frequencies across all reviewed protein sequences, representing the broadest available view of protein sequence space.
- **PDB X-ray** (~2025): frequencies from a 10,000-entity sample of X-ray crystallographic structures in the RCSB Protein Data Bank.

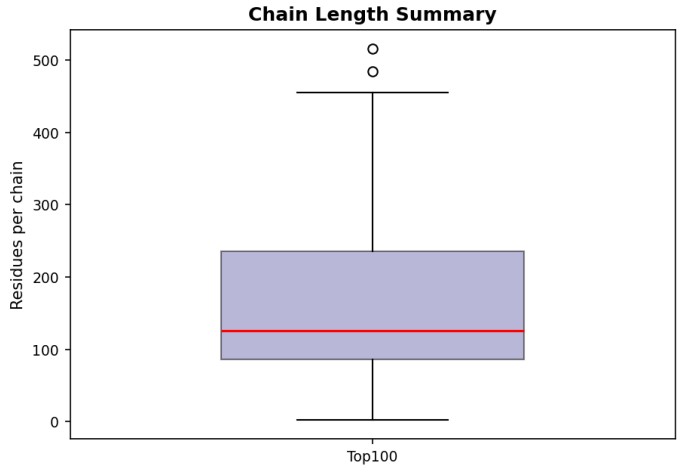
The two reference distributions are highly similar (most amino acids differ by <0.5%), indicating that the PDB's well-known crystallization bias — overrepresentation of soluble, globular, well-expressing proteins from model organisms — has only a modest effect on overall amino acid composition. The largest reference-to-reference differences are: Leu (9.66% UniProt vs 8.61% PDB, reflecting underrepresentation of membrane proteins), Gly (7.07% vs 7.69%, reflecting enrichment of compact folds with tight turns), and Arg (5.53% vs 4.83%, flexible sidechains harder to resolve crystallographically).

The Top100 dataset (17,434 residues from 98 structures) shows additional deviations from both references due to:

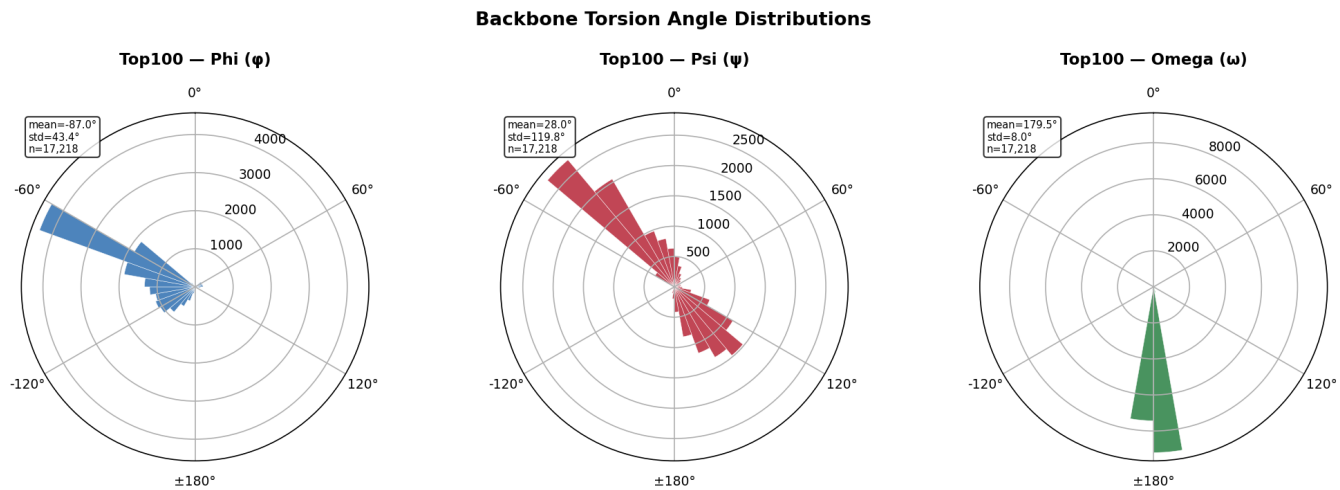
1. **Quality filtering** — high-resolution, low-B-factor selection favors well-ordered regions and stable folds, which can shift composition toward residues prevalent in secondary structure cores (Ala, Val) and away from surface-exposed flexible residues (Arg, Lys, Glu).
2. **Homology reduction** — filtering to ≤70% sequence identity reduces redundancy toward more uniform sampling of fold space, which can shift composition away from overrepresented protein families.
3. **Sample size** — with 98 structures, sampling variance is appreciable. The larger top2018 datasets (11,000+ structures) converge substantially closer to the PDB reference.

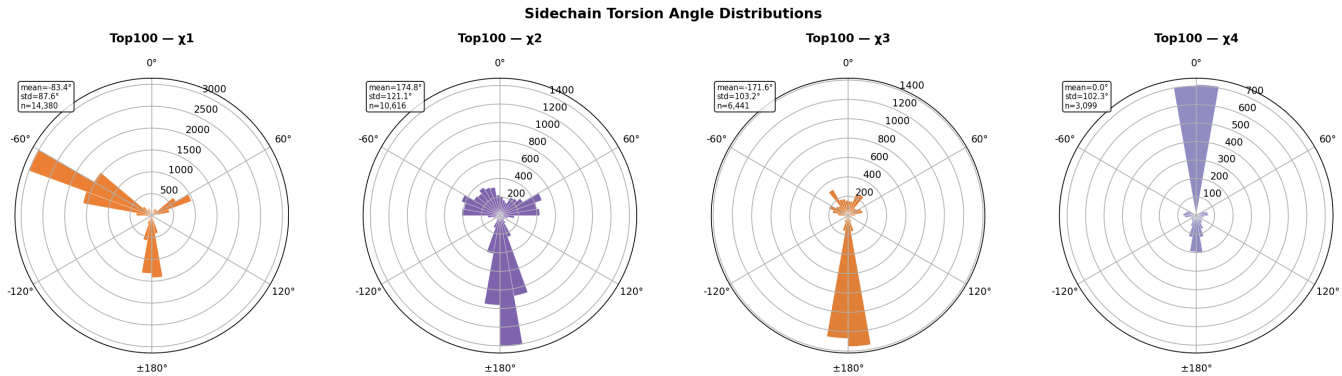
The Δ UniProt column quantifies the per-residue deviation from the sequence universe baseline.

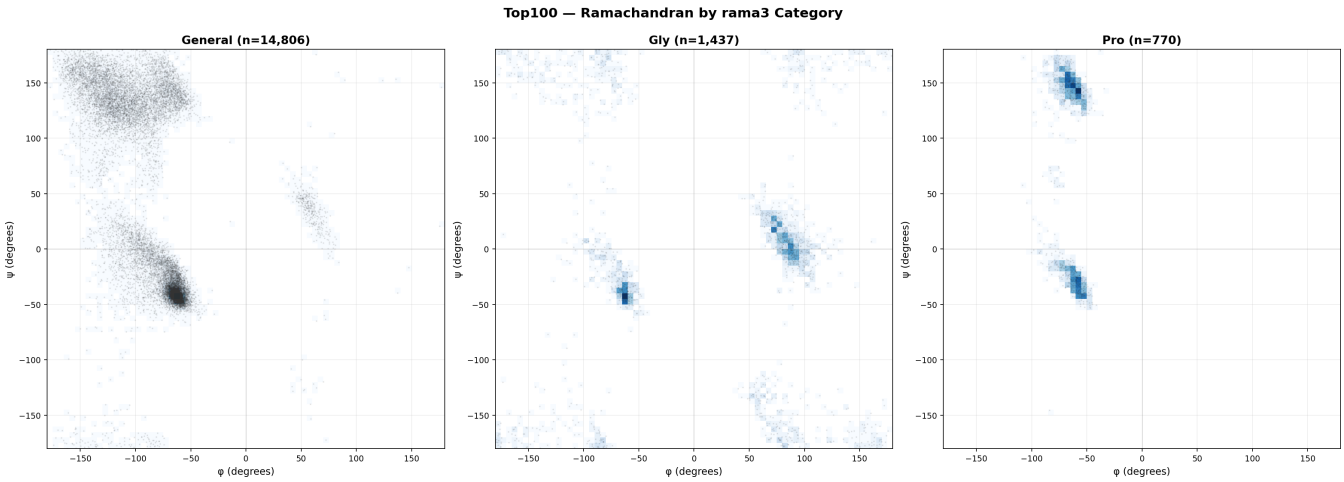





```
Chain length statistics (106 chains):
  Min:              2
  Q1:              86
  Median:         126
  Mean:         164.5
  Q3:             235
  Max:            516
  Std:          119.8
```





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Category counts (rama3):
  General        14,806  (84.93%)
  Gly             1,437  ( 8.24%)
  Pro               770  ( 4.42%)
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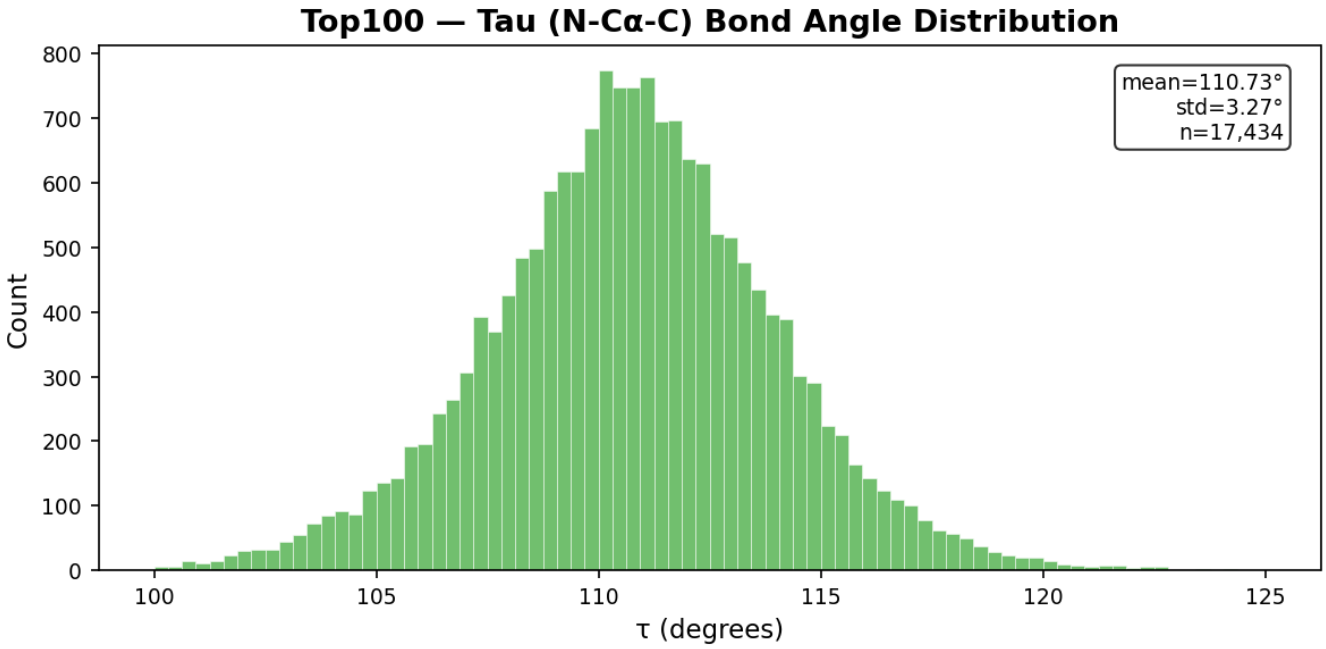

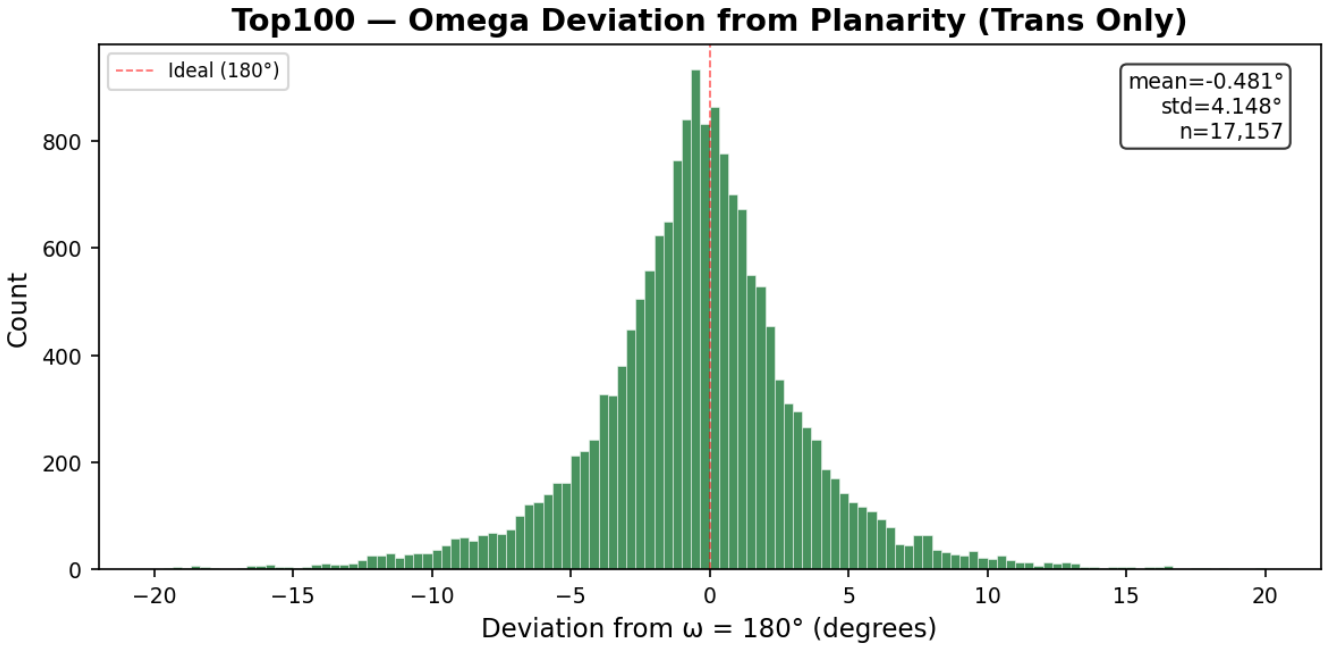


```
Dataset:                    Top100
Total residues:             17,434
Unique structures:          98
Unique chains:              106

Ramachandran category distribution:
  General         12,187  (69.90%)
  IleVal           2,052  (11.77%)
  Gly              1,437  ( 8.24%)
  TransPro           720  ( 4.13%)
  PrePro             567  ( 3.25%)
  CisPro              50  ( 0.29%)
```

Angle	N	Circ Mean	Circ Std
phi	17,218	-86.970	43.398
psi	17,218	27.990	119.824
omega	17,218	179.521	7.966
tau	17,434	110.729	3.269
chi1	14,380	-83.355	87.557
chi2	10,616	174.761	121.133
chi3	6,441	-171.584	103.209
chi4	3,099	0.024	102.314